MILITARY SPECIFICATION

CARTRIDGE, 60MM, ILLUMINATING, XM721 LOADING, ASSEMBLING, AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This Specification covers requirements and quality assurance provisions for the loading, assembling, and packing for one type of mortar cartridge designated as Cartridge, 60MM, Illuminating, XM721.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-286 - Propellants, Solids: Sampling, Examination and Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1310

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MIL-STD-414 - Sampling Procedures and Tables For Inspection by Variables for Percent Defective
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, and ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9345338 - Cartridge, 60MM, Illuminating, XM721 W/Fuze, Time XM776
9349695 - Container, Ammunition, Fiber, For Cartridge, 60MM, Illuminating, XM721
9354466 - Container, Ammunition, Metal For Cartridge, 60MM, Illuminating, XM721
9354467 - Box, Wirebound
9390710 - Packing and Marking for Container, Ammunition, Fiber for Cartridge, 60MM, Illuminating, XM721
9390712 - Packing and Marking For Container, Ammunition, Metal For Cartridge, 60MM, Illum, XM721
9390711 - Packing and Marking For Box, Wirebound, Packing Ammunition For Cartridge, 60MM, Illuminating, XM721

INSPECTION EQUIPMENT DRAWINGS

9201136 Tunnel, Light
9201268 - Procedure, Light Output Measurement
9201390 - Procedure, Photocell Checkout
9247071 - Photocell
9200964 - Photometer Head With Cell
7297287 - Chamber Gage
7257965 - Limit Check Gage

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications, standards and this specification.

3.2 Cartridge. The cartridge shall comply with all requirements on drawing (dwg.) 9345338 and associated dwgs. and with all requirements specified in applicable specifications standards and this specification.

3.3 Moisture.

3.3.1 First fire composition. The moisture content of the first fire shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.1 or 4.4.5.3.3.

3.3.2 Illuminant composition. The moisture content of the illuminant composition shall not exceed 0.30 percent, when tested as specified in 4.4.5.3.2 or 4.4.5.3.3.

3.4 Static functioning of the illuminant assembly. The average luminosity of the illuminant candle assembles shall be not less than 350,000 candlepower. The burn time shall be not less than 40 seconds. Determination shall be as specified in 4.4.5.4 (see 6.8).

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.4.5.7.

3.5.1 Misfire. The cartridge shall not fail to fire from the weapon.

3.5.2 Fuze/separation. The fuze shall function and the tail cone shall separate from the body tube assembly.

3.5.3 Functioning. The cartridge shall function satisfactorily (see 6.5) and shall provide effective illumination (see 6.6) for 32 seconds, min.

3.5.4 Security of parts. There shall be no metal parts failure in the bore of the weapon or in flight.

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3.5.5. Horizontal range. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.5.6 Fuze function time. The fuze assemblies shall function and shall comply with the fuze function time in Table I.

TABLE I Fuze Setting

Fuze setting (seconds)	Charge	Fuze function time (seconds)
19.0	1	19.0 ± 1.0
26.0	4	26.0 ± 1.0

3.5.7 Premature burst. The fuze shall not function in the bore of the weapon or in the first 100 meters of flight.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.

3.7.1 Parts and assemblies. All parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material that may adversely affect fit or function. The cleaning method used shall not be injurious to the parts nor shall any parts be contaminated.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Cloth. All cloth shall be free of tears, rips, and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears, or foreign matter.

3.7.4 Painting and marking. The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a past of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Alternate inspection provisions. Alternate inspection procedure, methods or equipment (including tool inspection, statistical peocess control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the procuring Contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative (s) is equal to or better then the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative (s) provide equal assurance, the provisions of this specification shall apply.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in the quantities indicated in Table XI.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078 and 4.4.3.5

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TABLE II First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70714 (AR)

PARAGRAPH	TITLE: Cartridge, 60mm, Illuminating M721		SHEET 1 OF 1		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fast fire composition (dwg. 12630532)	4oz	100%	3.3.1	4.4.3.1.1/4.4.5.3.1
	Illuminant composition (dwg. 12630532)	4oz	100%	3.3.2	4.4.3.1.2/4.4.5.3.2
	Static functioning (dwg. 12630532)	20	100%	3.4	4.4.3.2/4.4.5.4
	Projectile loading assembly (dwg. 9345340)				
	Examination for defects	30	100%	3.2	4.4.2.2/4.4.2.3
	Cartridge (dwg. 9345338)				
	Examination for defects	50*	100%	3.2	4.4.2.4/4.4.2.5
	Proving Ground	50*	100%	3.5	4.4.3.3/4.4.5.7

NOTES: *Same group of samples

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot of Cartridges 60mm, Illuminating shall contain:

a. Projectile body metal parts in accordance with the same drawing,same drawing revision, same specification, and same specification revision, of one interfix number from one manufacturer.

b. Ignition cartridges of one lot interfix number from one manufacturer.

c. Fin assemblies of one lot interfix from one manufacturer.

d. Propelling charges of one lot interfix number from one manufacturer.

e. Loaded fuzes of one interfix from one manufacturer.

f. Parachute assemblies from one lot interfix number from one manufacturer.

g. Canister assemblies from one manufacturer in accordance with the same drawing and same drawing revision.

h. Obturating rings of one lot interfix from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70714 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Canister Loading Assembly		12630532
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	First fire composition test	(a)		3.3.1	4.4.5.3.1
102	Illuminant composition test	(b)		3.3.2	4.4.5.3.2
103	Ignition disc assembly missing		0.40%	3.2	Visual
104	Filler rings not flush to 1/16 inch above flush		0.40%	3.2	Visual
105	Swivel fails to rotate freely		0.40%	3.2	Visual/Manual
106	Static functioning test	(c)		3.4	4.4.5.4
107	Tape over filler rings missing, damaged or improper		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
a. Sample per 4.4.3.1.1
b. Sample per 4.4.3.1.2
c. Sample per 4.4.3.2

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Projectile, 60mm: Illuminating XM721 Loading Assembly During Assembly Operation		9345340
			NEXT HIGHER ASSEMBLY 9345338

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Wire loop assembly pull test		(a)	3.2	Gage
102	Parachute cup missing		0.40%	3.2	Visual
103	Baffle disc missing		0.40%	3.2	Visual
104	Cable protector missing or overlapped		0.40%	3.2	Visual
105	Shroud lines pulled out of parachute bag		0.40%	3.2	Visual
106	Parachute bag closure cord not removed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
(a) See 4.4.3.3

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Projectile, 60mm Illuminating XM721 Loading Assembly	1	1	9345340
				NEXT HIGHER ASSEMBLY 9345338

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of shear pin holes		0.40%(a)	3.2	Gage
102	Diameter of shear pin holes		0.40%(a)	3.2	Gage
103	Location of shear pin holes		0.40%	3.2	Gage
104	Shear pins missing or loose		0.40%	3.2	Visual
105	Shear pins not flush or below		0.40%	3.2	Visual
106	Tail cone not fully seated		0.40%	3.2	Visual
107	Shear pins not staked		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
(a) Alternate sampling plan may be usd as specified in 4.4.3.4

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cartridge, 60mm Illuminating XM721, Prior to Assembly of Fuze, Fin Assembly & Ignition Cartridge	1 OF 1	9345338
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Damaged threads on fin boom, tail cone, fuze, or fuze adapter		0.40%	3.2	Visual
102	Damaged threads on ignition cartridge or rear of fin		0.40%	3.2	Visual
103	Adhesive/sealant missing where required		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cartridge, 60mm Illuminating XM721 W/Fuze Time, XM776	1 OF 2	9345338
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Ignition cartridge missing,cracked or damaged		100%	3.2	Visual
b	Propelling charge increment(s) missing, cracked or damaged		100%	3.2	Visual
c	Fuze safety wire missing		100%	3.2	Visual
d	Fuze not set on safe		100%	3.2	Visual
e	Obturator band missing		100%	3.2	Visual
Major					
101	Assembly fails to freely traverse alignment gage for 20 inches, min.		100%	3.2	Gage
102	Inadequate torque applied to fin boom		0.40%	3.2	Torque test/4.4.5.5
103	Inadequate torque applied to fuze		0.40%	3.2	Torque test/4.4.5.5
104	Damaged fin blades		0.40%	3.2	Visual
105	Security of parts		0.40%	3.2	Manual/4.4.5.5
106	Total length, max.		0.40%	3.2	Gage
107	Inadequate torque applied to ignition cartridge		0.40%	3.2	Torque test/4.4.5.5
108	Obturator band not flush or below flush		100%	3.2	Visual
109	Tack weld missing		100%	3.2	Visual
Minor					
201	Protective finish damaged		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cartridge, 60MM Illuminating XM721 W/Fuze Time, XM776 (Cont)	2 OF 2	9345338
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
202	Markings incorrect, illegible or missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual
	PROVING GROUND TEST				
	Ballistic Acceptance	50		3.5	4.4.3.5/4.4.5.7

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Fiber Container (Unsealed)	1 OF 1	9390710
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Propelling charge support assembly missing		100%	5.1	Visual
b	Support assembly position improper		100%	5.1	Visual
Major					
101	Support nose missing		0.40%	5.1	Visual
102	Cushion missing		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Fiber Container (Sealed)	1 OF 1	9390710
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged		0.40%	5.1	Visual
102	Sealing strip incomplete, badly wrinkled or missing		0.40%	5.1	Visual
103	Contents loose		0.40%	5.1	Manual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Metal Ammunition Contained (Unsealed)	1	1	9390712
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any material containing surface moisture or condensate		0.40%	5.2	Visual
102	Spacer missing		0.40%	5.2	Visual
103	Cushion missing		0.40%	5.2	Visual
Minor	None defined				

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.9	TITLE Metal Ammunition Contained (Sealed)		SHEET 1 OF 1		DRAWING NUMBER 9390712
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Marking missing, misleading or unidentifiable		0.40%	5.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing.

PRECAUTION: This Specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Moisture content.

4.4.3.1.1 First fire composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.1.2 Illuminant composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.2 Static functioning. The following sampling plan will apply with the sample tested as specified in 4.4.5.4.

4.4.3.2.1 Sampling. Each inspection lot will consist of approximately eighty-five candles produced consecutively by the mixing/pressing operation and will be designated as one batch (see 6.7) for identification purposes.

4.4.3.2.1.1 Sampling. Select a sample of three candles from each inspection lot (approximately 85 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements.

4.4.3.2.1.2 Sampling. After successful completion of 4.4.3.2.1.1, the sampling frequency may be reduced to three candles from each group of six inspection lots (approximately 510 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.1.1.

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4.4.3.2.1.3 Sampling. After successful completion of 4.4.3.2.1.2, the sampling frequency may be reduced to three candles from each group of inspection lots (approximately 1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.4. Continue this sampling until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event the one lot fails to comply with the requirements, the sampling shall revert to 4.3.3.2.1.2.

4.4.3.2.1.4 Sampling. After successful completion of 4.4.3.2.1.3, the sampling frequency may be reduced to three candles from eights hours of production or approximately 1615 candles, whichever is less. The canister assembly shall be observed for compliance with the requirements specified in 3.4. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.2.1.3.

4.4.3.2.1.5 Sampling. In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.2.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.2.1.1. If the continous mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.4.3.2.1.2.

4.4.3.2.1.6 Sampling. In the event of failure of any sample to comply with the requirements, each individual inspection lot within-the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

4.4.3.3 Wire loop assembly pull test. Eight (8) representative samples of the crimped oval sleeve to wire shall be pull tested, two (2) at the beginning of an 8 hour shift, four (4) at the midway of the shift, and two (2) at the end of the shift. Two (2) samples shall also be selected immediately following any adjustment to the machine which affects the crimping load regardless of when the adjustment is made. The test shall be performed as specified in 4.4.5. Failure of any sample to withstand the required pull force shall be cause for rejection of previous four (4) hours production made on the applicable crimping machine until it is requalified by 100 percent screening of the rejected production.

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4.4.3.4. Alternate sampling plan for examination of depth and diameter of rear shear pin holes. The depth and diameter of the shear pin holes shall be inspected in the following manner using approved gages.

4.4.3.4.1 First production shift examination. Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every 1/2 hour until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free.

4.4.3.4.2 Second production shift examination. After successfully completing the sampling of 4.4.3.4.1 for one eight hour production shift with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every one hour and at the beginning and at the end of the shift. If a defect is found the sampling plan of 4.4.3.4.1 will be reinstated.

4.4.3.4.3 Third production shift examination. After successfully completing the sampling plan of paragraph 4.4.3.4.2 for one eight hour production shift with no defects the sampling plan may be reduced to three (3) drilled units being inspected every hour and at the beginning and end of the shift. If a defect is found, the sampling plan of 4.4.3.4.2 will be reinstated.

4.4.3.4.4 Fourth production shift examination. After successfully completing the sampling plan of paragraph 4.4.3.4.3 for one eight hour shift with no defects, the sampling plan may be reduced to two (2) drilled units being inspected every two hours. If a defect is found, the sampling plan of 4.4.3.4.3 will be reinstated.

4.4.3.4.5 Tooling change. A change of tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change.

4.4.3.4.6 Production lapse. A stop in production of over two months will require starting the inspection at the first production shift plan.

4.4.3.5 Proving ground. Fifty sample cartridges shall be selected for this test. The test shall be performed as specified in 4.4.5.7. Defect classification and individual accept/reject criteria shall be in accordance with the following table.

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Categories	Defect	Accept/Reject	
Critical:	Premature burst or metal parts separation	0	1
Major:	Misfire	1	2
Major:	Fuze non-function or non-separation of tail cone from body tube assembly	1	2
Major:	Horizontal range less than 80% of average range	0	1
Major:	Fuze functioning time (See Note 1)	See 4.4.3.5.1.1	
Major:	Illuminant fails to ignite or provide effective illumination for 32 seconds or less (See Note 2)	2	3

Note 1: If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions properly in all respects with the exception that the "effective illumination" time is less than thirty-two seconds because the illuminant and parachute assembly hit the ground while still burning, the illumination time defects shall not be additionally assessed against that round.

Note 2: If a cartridge functions properly in all respects with the exception that the "effective illumination" time is less than thirty-two seconds because the illuminant and parachute assembly hit the ground while still burning and if the burst height, that is, the height above ground level at which the tail cone separated from the tube body was less than 300 meters the performance of the cartridge will not be considered in evaluating the acceptability of the lot for illumination time defects.

4.4.3.5.1 Overall acceptance criteria. In addition to the individual accept/reject criteria of 4.4.3.5, the lot shall be rejected if a total of three or more Major defects occur.

4.4.3.5.1.1 Acceptance criteria for fuze. The lot shall also be rejected if the percent lot defective for fuze functioning time are computed with the use of MIL-STD-414, Section B, with an AQL of 1.5 percent, exceeds the maximum allowable percent lot defective for each setting.

4.4.3.6 Strength of heat weld. One obturating ring sample, minimum, after every 100 projectiles, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.4.5.8. Failure of the sample to meet the requirement shall be cause for those 100 preceding projectiles represented by the sample to be rebanded and retested.

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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the table starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Methods of inspection.

4.4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing, specification or specification requirements, and that all specified manufacturing processes have been followed.

4.4.5.2 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3, 4, and 5 of the specification in accordance with 4.4.2 herein.

4.4.5.3 Moisture.

4.4.5.3.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of dimethylformamidemethyl alcohol and anhydrous methanol.

4.4.5.3.2 Illuminant composition. The moisture content of illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in MIL-STD-1234, Method 101.2 up to paragraph 5.3. A 50 ml. sample of 1:1 dimethylformamidemethyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix the solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes. A blank sample of 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute.

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The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.50 - 0.25 minutes. 25 ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker throught the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as specified above for the extraction of water from the sample . The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation: $\frac{AB}{5\ (10)}$ = Percent H_2O

Less _____ Std. Blank

Leaves _____ Percent Moistur in Sample

Where:

A = mls of Karl Fischer to titrate sample

B = Standardization Factor (mg./ml.)

10 = Constant Factor

5 = Standard Sample Weight

4.4.5.3.3 Alternate method for first fire composition moisture and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4, MIL-STD-286. The temperature setting shall be 100 degrees Centigrade and the time setting 30 minutes. The sample size shall be between 0.40 and 0.60 grams.

4.4.5.4 Static functioning test. The test equipment required for this test is delineated in 2.1.2 under Inspection Equipment Drawings. The temperature of candle shall be 700 ± 5°F at time of test. The average candlepower is to be calculated over the best continuous 32 seconds of illuminant readings. The first fire disc assemblies shall be removed prior to testing. The distance from the photometric cell to the flare shell shall be 60.00 ± .25 feet. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is 14-15 feet. The center line of baffle opening shall be in line with axial center of fireball and axial center of photometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on ash buildup the tunnel hearth shall be repainted at least once in a three

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month duration to maintain the black interior or more frequently as determined by the resident Government Quality Assurance Representative. If the individual burning time or average candlepower does not meet the requirement, the illuminant inspection lot represented by the sample shall be rejected.

4.4.5.5 Torque test. The samples supplied in accordance with 4.4.2.5 shall be tested by applying the required assembly torque to the components in the direction of assembly before the thread sealing compound has set. Any perceptible movement of parts will be considered as failure of the assembly to comply with the requirement.

4.4.5.6 Wire loop assembly pull test. The test sample shall be assembled on a universal testing machine and the load applied in an axial direction until the required pull force is exerted. The rate of travel of the machine shall be 2 inches per minute maximum. Samples subjected to the pull test shall be considered as destructive test samples.

4.4.5.7 Proving ground. The test shall be performed at a Government Proving Ground. All the cartridges shall be fired from a 60MM Mortar for which the round is standard. The cartridge shall be assembled with accepted fuzes, fin assemblies, ignition cartridges, primers, and propellant increments. One half of the sample cartridges shall be fired with one propelling charge increment (charge zone one) and one half of the sample cartridges shall be fired with four propelling charge increments (charge zone four) and in accordance with Table III. However, if meteorological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to increase the height of burst and insure effective illumination. Observation shall be made for compliance with the requirements specified for security of parts, functioning, functioning time, horizontal range and "effective illumination" (See 6.6). Fuze functioning time shall be reported to al least the nearest .01 sec, illuminant burn time to the nearest 0.5 second, and horizontal range on the ground to a point under the position of fuze functioning to the nearest 5 meters. The cartridges shall be conditioned at 75 plus or minus 5 degrees Fahrenheit for a minimum of 4 hours prior to testing and fired at this temperature. Any malfunction occurring during the test shall be reported in the official firing record. Meteorological data shall also be recorded in the official firing record. Muzzle velocity shall also be recorded for each round fired and reported in the official firing record.

TABLE III TEST SETTINGS

Charge Zone	Fuze Setting (seconds)	Mortar Elevation (degrees)
1	19.0	85
4	26.0	45

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4.4.5.8 Strength of heat weld. The sample shall be tested for integrity by applying the required load at the heat weld radially and in a direction away from the axis (center) of the ring.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridge shall be in accordance with the requirements of drawing 9390710.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Same as level A.

5.2 Packing

5.2.1 Level A. Packing for cartridge shall be in accordance with the requirements of drawing 9390712.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Same as level A.

5.3 Marking.

5.3.1 Level A. Unit packaging for cartridge shall be marked in accordance with the requirements of drawing 9390710. Packing shall be marked in accordance with the requirements of drawing 9390712.

5.3.2 Level B. Same as level A.

5.3.3 Level C. Same as level A.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly to M721 illuminating cartridges for use with the 60mm mortar (LWCMS).

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, ARRADCOM, Edgewood, Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.4 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army ARDEC, ATTN: AMSMC-QAT-I (D), Bldg 62, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.5 Satisfactory functioning. Satisfactory functioning shall be defined as ejection of the parachute and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination".

6.6 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.7 Illuminant inspection lot. An illumination inspection lot is defined as the quantity of material which has been manufactured at one time by some uniform chemical process using the same lot of components or materials. The inspection lot of candles shall be defined as that quantity of candles which has been loaded with one inspection lot of illuminant using the one lot of illuminant for all increments.

6.8 Static burning time. Minimum static burning time as referenced herein (see 3.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee meeting the final ballistic minimum burning time requirement. Structural integrity of the candle under set back loading is a prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

6.9 Subject term (key word) listing.
First fire composition
Illuminant composition
Moisture
Static functioning

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A431)

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MIL-C-70714	CARTRIDGE, 60MM, ILLUMINATING, XM721 LOADING, ASSEMBLING, AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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